

A Unit-Norm Full-Spark Frame That Is Not Splittable

Counterexample to the full-spark splittability conjecture

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Status. Candidate full counterexample, likely valid, for human review. The proof is exact and finite-dimensional; novelty remains subject to broader literature review.

1 Source conjecture

For a finite frame $F = (f_1, \dots, f_n)$ spanning $\mathbb{R}^d$, define

$$\|z\|_F = \min\{\|q\|_1 : Fq = z\}, \quad \Sigma_{F,s} = \bigcup_{|S| \leq s} \text{span}\{f_j : j \in S\}.$$

Choose a best s -term approximant

$$z_s \in \operatorname{argmin}_{v \in \Sigma_{F,s}} \|z - v\|_F.$$

Chen, Kummerle, and Wang call F *s-splittable* with constant $\beta > 0$ if, for every $x, y \in \mathbb{R}^d$,

$$\|x + y\|_F \geq \|x_s\|_F - \|y_s\|_F + \beta(\|y - y_s\|_F - \|x - x_s\|_F). \quad (1)$$

Immediately after the definition they conjecture that every full-spark frame is splittable for a sufficiently small positive β [1]. The construction below disproves this already for $d = 2$ and $s = 1$.

Remark 3.4. [30, Theorem 7.2] states that the following three conditions are equivalent if F is full spark (every d columns of F are linearly independent) [6]:

- (i) A has F -NSP- k .
- (ii) A has F -SNSP- (k, c) for some $c > 0$.
- (iii) AF has NSP- k .

Theorem 3.5 ([30, 29]). $\Delta_{F,A,0}(Az) = z$ for any $z \in \Sigma_{F,k}$ if and only if A satisfies F -NSP- k .

Theorem 3.5 is analogous to Theorem 2.3. For a result such as Theorem 2.5, we need the frame F to be somewhat regular in the following way:

Definition 3.6 ([29, Definition 2.5]). We call F s -splittable with constant $\beta > 0$ if for any $x, y \in \mathbb{R}^d$,

$$(3.3) \quad \|x + y\|_F \geq \|x_s\|_F - \|y_s\|_F + \beta(\|y - y_s\|_F - \|x - x_s\|_F).$$

Requiring F to be regular in the above way is not surprising as the unit ball of F -norm (1.3) is the convex hull of columns in F (together with its negatives). In the case where F is the canonical basis, we have $\beta = 1$, which is the optimal value. We conjecture that all full-spark frames are splittable with small enough β , and it would be interesting work to estimate β based on certain frame properties of F .

Theorem 3.7 ([29, Corollary 5.3]). Let F be s -splittable with constant β and A has F -RNSP- (k, ρ, τ) with $\rho < \beta$, then for any $z \in \mathbb{R}^d$ and $\|w\|_2 \leq \eta$, we have

$$\|z - \Delta_{F,A,\eta}(Az + w)\|_F \leq \frac{(1+\beta)(1+\rho)}{\beta(\beta-\rho)} \sigma_{F,s}(z) + \frac{2\tau(1+\beta)}{\beta-\rho} \eta.$$

The above theorem is a generalization of (2.5) in Theorem 2.5 (with $q = 1$). It remains to be future work to establish a performance guarantee like (2.6) which should involve a modified

Definition 3.6 and the full-spark splittability conjecture, source PDF page 7.

2 Idea of the proof

With three atoms in the plane, the atomic unit ball is a centrally symmetric hexagon, so both its norm and the distance to each atom line have exact piecewise-linear formulas. Every pair (x, y) imposes a linear constraint $L \geq \beta D$ on the purported splitting constant. We choose rational unit atoms and two rational witness pairs: the first has $L, D < 0$ and forces a strict lower bound on β , while the second has $L, D > 0$ and forces a smaller upper bound. Strict distance comparisons and exposed facets make all four best one-term approximants unique.

3 Counterexample and exact atomic norm

Set

$$a = \left(-\frac{7}{25}, -\frac{24}{25}\right), \quad b = (1, 0), \quad c = \left(\frac{20}{29}, -\frac{21}{29}\right), \quad F = (a, b, c). \quad (2)$$

Theorem 1. The frame in (2) is a unit-norm full-spark frame for $\mathbb{R}^2$, but it is not 1-splittable for any positive constant β . In fact, no real β satisfies (1) for all x, y .

Each column has Euclidean norm one, and

$$\det(a, b) = \frac{24}{25}, \quad \det(a, c) = \frac{627}{725}, \quad \det(b, c) = -\frac{21}{29}.$$

Thus every pair is linearly independent, so F is full spark.

Write $N(z) = \|z\|_F$. Its unit ball is the centrally symmetric hexagon $\text{conv}\{\pm a, \pm b, \pm c\}$. The three pairs of opposite supporting lines give the exact formula

$$N(z_1, z_2) = \max \left\{ \left| z_1 + \frac{3}{4}z_2 \right|, \left| -\frac{3}{11}z_1 + \frac{37}{33}z_2 \right|, \left| -z_1 + \frac{3}{7}z_2 \right| \right\}. \quad (3)$$

Indeed, the first functional takes the values $-1, 1, 17/116$ on a, b, c ; the second takes $-1, -3/11, -1$; and the third takes $-23/175, -1, -1$. Hence each has absolute value at most one on the six vertices and exposes the claimed facet. These are all three pairs of facets of the hexagon, proving (3).

4 Best one-term approximation

For $v \in \{a, b, c\}$, let

$$d_v(z) = \min_{t \in \mathbb{R}} N(z - tv).$$

Lemma 1. *The distances to the three atom lines are*

$$d_a(z) = \frac{|24z_1 - 7z_2|}{24}, \quad (4)$$

$$d_b(z) = \frac{25|z_2|}{24}, \quad (5)$$

$$d_c(z) = \frac{25|21z_1 + 20z_2|}{627}. \quad (6)$$

Whenever one of these values is strictly smaller than the other two, the best one-term approximant is unique.

Proof. The annihilator $(24, -7)$ of a has dual norm

$$\max\{|(24, -7)a|, |(24, -7)b|, |(24, -7)c|\} = 24,$$

with the maximum attained only at $\pm b$. The quotient norm modulo $\text{span}(a)$ is therefore (4). Equality can occur only when the residual is a multiple of b , which determines the minimizing point on $\text{span}(a)$ uniquely. Similarly, $(0, 1)$ annihilates b , has dual norm $24/25$ attained only at $\pm a$, and gives (5). Finally, $(21, 20)$ annihilates c , has dual norm $627/25$ attained only at $\pm a$, and gives (6). Thus the minimizing point on each selected line is unique. Strict inequality between the three distances also makes the selected atom line unique. $\square$

5 Two incompatible constraints on the splitting constant

Take first

$$x = \left(-\frac{4}{5}, \frac{3}{5}\right), \quad y = \left(\frac{7}{15}, -\frac{8}{15}\right).$$

Lemma 1 gives

$$(d_a(x), d_b(x), d_c(x)) = \left(\frac{39}{40}, \frac{5}{8}, \frac{40}{209}\right),$$

and

$$(d_a(y), d_b(y), d_c(y)) = \left(\frac{28}{45}, \frac{5}{9}, \frac{65}{1881} \right).$$

Thus both best one-term approximants are unique and lie on $\text{span}(c)$. They and their residuals are

$$\begin{aligned} x_1 &= -\frac{1131}{1045}c = \left(-\frac{156}{209}, \frac{819}{1045} \right), & x - x_1 &= \frac{40}{209}a, \\ y_1 &= \frac{6496}{9405}c = \left(\frac{896}{1881}, -\frac{1568}{3135} \right), & y - y_1 &= \frac{65}{1881}a. \end{aligned}$$

Formula (3) now yields

$$N(x_1) = \frac{1131}{1045}, \quad N(y_1) = \frac{6496}{9405}, \quad N(x + y) = N\left(-\frac{1}{3}, \frac{1}{15}\right) = \frac{38}{105}.$$

After moving the non- β terms in (1) to the left, this witness gives

$$\underbrace{N(x + y) - N(x_1) + N(y_1)}_{-391/13167} \geq \beta \underbrace{(N(y - y_1) - N(x - x_1))}_{-295/1881}. \quad (7)$$

Both quantities are negative, so (7) forces

$$\beta \geq \frac{391}{2065}. \quad (8)$$

For the second witness, take

$$x' = b = (1, 0), \quad y' = \left(-\frac{3}{10}, -\frac{1}{5}\right).$$

The best approximant to x' is uniquely $x'_1 = b$. For y' the three distances are

$$(d_a(y'), d_b(y'), d_c(y')) = \left(\frac{29}{120}, \frac{5}{24}, \frac{515}{1254}\right),$$

so its unique best approximant lies on $\text{span}(b)$, with

$$y'_1 = \left(-\frac{29}{120}, 0\right), \quad y' - y'_1 = \frac{5}{24}a.$$

Moreover

$$N(x' + y') = N\left(\frac{7}{10}, -\frac{1}{5}\right) = \frac{11}{14}.$$

Consequently (1) gives

$$\underbrace{\frac{11}{14} - 1 + \frac{29}{120}}_{23/840} \geq \beta \underbrace{\frac{5}{24}}_{N(y' - y'_1) - N(x' - x'_1)}, \quad (9)$$

which forces

$$\beta \leq \frac{23}{175}. \quad (10)$$

But

$$\frac{391}{2065} - \frac{23}{175} = \frac{598}{10325} > 0.$$

The necessary bounds (8) and (10) are incompatible. This proves Theorem 1.

6 Verification, scope, and novelty status

All displayed values are rational. The accompanying standard-library script `verify_exact.py` checks the minors, facet inequalities, distance comparisons, decompositions, norms, and final gap using exact rational arithmetic. It is an audit aid; the proof above is self-contained.

The general definition can involve nonunique best sparse approximants [2]. Every best approximant used here is unique, so the counterexample does not depend on how an argmin is selected. Unit-norm columns also show that the obstruction is not caused by rescaling a frame vector.

A bounded search of the run indexes and exact web queries found the source conjecture and the earlier atomic-norm article, but no later proof or counterexample. Priority is not asserted. The result should receive expert proof review and a broader citation search before dissemination.

References

- [1] X. Chen, C. Kummerle, and R. Wang, *Sparse Recovery for Overcomplete Frames: Sensing Matrices and Recovery Guarantees*, arXiv:2408.16166 (2024).
- [2] X. Chen, *A Unified Recovery of Structured Signals Using Atomic Norm*, Information and Inference 13 (2024), iaad050; arXiv:2207.06484.